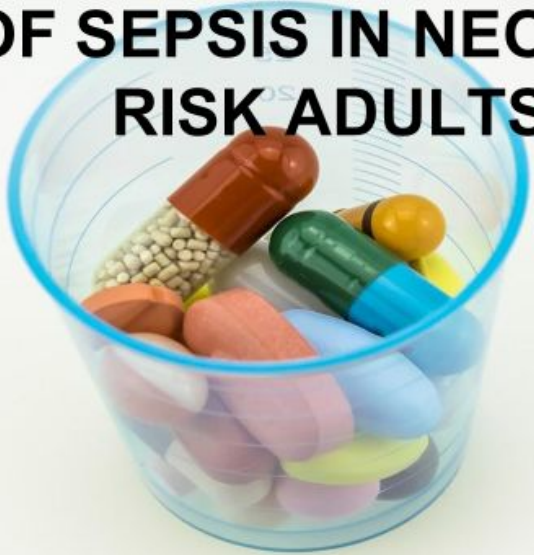
INTELLIGENT SENSOR SYSTEM FOR EARLY PREDICTION OF SEPSIS IN NEONATES AND HIGH RISK ADULTS



Progress Review meeting January 2026

Literature Survey on Bacterial-Specific Biomarkers for SERS-Based Sepsis Detection:

- Comprehensive systematic review of research publications , analyzing bacterial-specific surface biomarkers with established Raman spectroscopic signatures and demonstrated SERS detectability. Critical evaluation of biomarkers superior to traditional host-response markers (PCT, CRP) for enabling rapid, bacterial-specific sepsis.

Collaborative Review Meetings with the Research Team:

- Multiple strategic review meetings were conducted with project team and research collaborators to discuss preliminary findings, obtain expert input on biomarker selection rationale, and identify optimization priorities.

Meetings resulted in:

- Confirmation of biomarker panel selection aligning with clinical sepsis diagnostic needs.
- Molecular simulations suggestions with optimization and frequency analysis and strategic planning for next phase approaches were carried out.

Finalization of Bacterial Biomarker Panel:

Focus Area

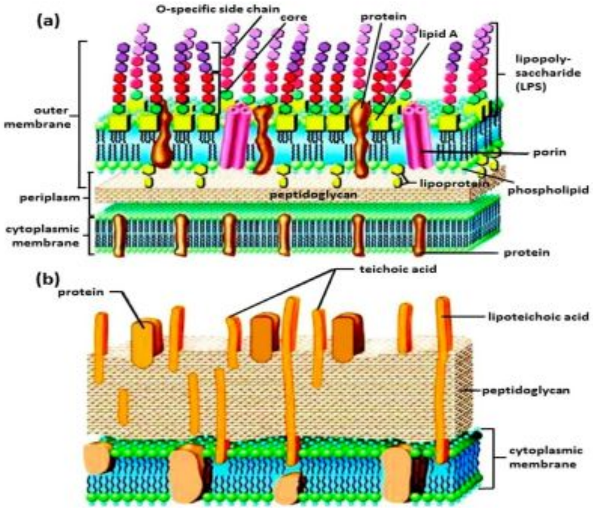
- Markers specific to the surface of bacteria and not overlapping with other pathogens / Readily available or expressible markers only in bacteria.
- Help healthcare professionals with confirmation of a bacterial infection and subsequent gram classification for early diagnosis and treatment and SERS based detection compatibility.

Rationale for biomarkers selection:

1.**Universal Bacterial Detection** :N-acetylmuramic acid

Exclusivity:

- Present in ALL bacteria, absent in viruses/fungi/parasites
- Direct bacterial confirmation
- Unique to bacterial peptidoglycan only



2. <u>**Gram classification**</u>

2.a **Gram-Positive–Specific Markers** : Lipoteichoic Acid (LTA)

<u>Uniqueness:</u>

- Only in Gram-positive bacteria , surface accessibility .

2b. **Gram Negative specific marker** :Lipopolysaccharides (Lipid A , O antigen)

<u>Uniqueness:</u>

- Only in Gram-negative bacteria, absent in all other organisms
- Strong enhancement, endotoxin correlation .

Established optimized three-component biomarker panel: (1)N-acetylmuramic acid (PGN) - universal bacterial detection marker present in all (2) Lipopolysaccharide (LPS) - Gram-negative specific marker (3) Lipoteichoic Acid (LTA) - Gram-positive specific marker.

- Panel selected based on biomarker exclusivity to bacteria, surface accessibility for SERS detection, and Gram classification capability.
- These markers can be supplemented with bacterial-specific components to enhance detection sensitivity in complex cases where signals might be obscured.

Optimization and Preliminary SERS Characterization with Test Molecules:

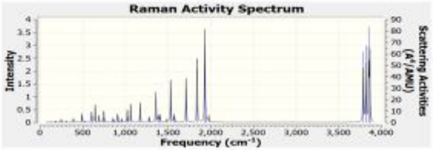
Computational Approach Overview :

To establish the theoretical foundation for Raman spectral characterization and validate biomarker detection feasibility, the team tried conducting comprehensive software-based molecular simulations and frequency analysis.

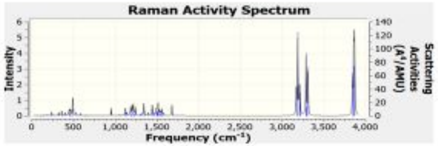
- The team conducted simulations using basic/test molecules relevant to biomarker analysis Uric Acid, Glucose, Creatinine, Lactic Acid ,Glycine, Urea etc.

Preliminary Outcomes:

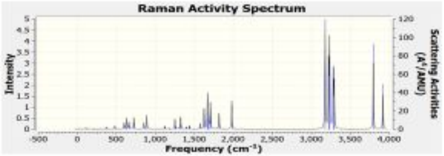
- Validation of spectral peak predictions against literature values
- distinct molecular fingerprints for different biomarkers



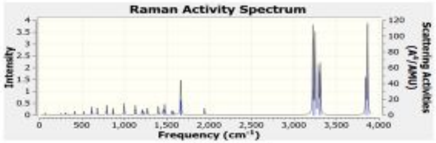
Uric Acid



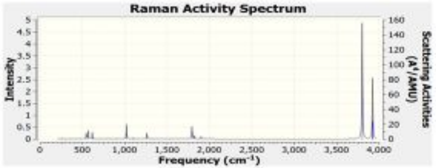
Glucose



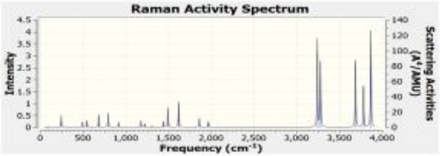
Creatine



Lactic acid



Urea



Glycine

Initiation of Pure Biomarker Sample Procurement Procedures

- Initiation of procurement procedures for commercially available high-purity bacterial biomarker standards: Trying to establish purchase with quality control specifications ensuring received biomarkers meet analytical standards for SERS characterization and clinical validation studies.

Planning and Strategy for Next Phase of Work

- Developing comprehensive roadmap for subsequent project phases, SERS substrate optimization with pure biomarker standards, reproducibility assessment across multiple substrate batches.

<u>Meetings and Review Sessions:</u>

Collaborative Review Meetings with project team throughout December 2025

Key Discussion Points:

- Strategic planning for next phase of project and translation roadmap.

Outcomes:

❖ Following comprehensive evaluation of project requirements, analytical standards, the research team finalized minimum order quantities for all three biomarker components

MOQ (minimum order quantity) Rationale:

To ensure cost efficiency and experimental continuity, vendors were engaged for complete vials. The following quantities allow for ~100 experiments per marker.

Justification for MOQ:

- ➔ NAM: 0.5-1 mg per experiment (100 mg = 100-200 experiments)
- ➔ LPS: 0.1 mg per experiment (10 mg = 100 experiments)
- ➔ LTA: 0.05 mg per experiment (5 mg = 100 experiments)

Biomarker	Supplier Form	MOQ	Unit	Est. Cost (INR)
N-Acetylmuramic Acid (NAM)	Analytical standard, lyophilized Purity: ≥98% (HPLC)	100	mg	₹12,000-20,000
Lipopolysaccharide (LPS, E. coli O111:B4)	Phenol extracted, Lyophilized powder Purity: ≥90-95%	10	mg	₹12,000-18,000
Lipoteichoic Acid(LTA, S. aureus, Gram-positive extract)	Purified antigen, solution/lyophilized powder Purity: >95%	10	mg	₹30,000-40,000
Total MOQ Procurement	₹54,000-78,000			
Contingency (15%)	₹12,000-20,000			
Total Recommended Budget	₹66,000-98,000			

Order Placement and Procurement

- Vendor Engagement: Successfully sourced multiple qualified suppliers through quotation process and placed order accordingly following administrative process of the institute.
- Receipt of the first contingent of bacterial biomarker standards and awaiting the second one.

Literature review of optimal SERS configuration for experimental phase:

- Literature review of biomarkers limits, solvents, capture probes and existing studies of NAM based on SERS technique.
- comprehensive review of recent scientific literature to establish the theoretical foundation for experimental work.

The project progress represents a critical transition point from planning and preparation to active experimental work. With procurement milestones and technical preparation, the research team is looking forward to execute the planned experimental work with confidence and efficiency.

Literature Review

LPS / endotoxin

Measurement of endotoxin activity in critically ill patients using whole blood neutrophil dependent chemiluminescence

John C Marshall [1,✉], Paul M Walker [2], Debra M Foster [3], David Harris [4], Melanie Ribeiro [4], Jeff Paice [5], Alexander D Romaschin [6], Anastasia N Derzko [7]

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PMCID: PMC125316 PMID: 12225611

EAA outperforms LAL for clinical endotoxin detection
The Endotoxin Activity Assay (EAA) enables rapid (<30 min), whole-blood detection of endotoxemia using <100 μL sample volume and shows high specificity for LPS (validated by polymyxin B inhibition), overcoming major limitations of the conventional LAL assay caused by circulating inhibitors and non-specific fungal interference.

Endotoxemia is a reliable diagnostic marker of Gram-negative sepsis
All patients with culture-proven Gram-negative infections exhibited endotoxin levels >50 pg/mL (100% sensitivity in this cohort), with significantly higher levels in sepsis compared to non-sepsis patients; endotoxemia correlated with leukocytosis but not mortality, indicating strong diagnostic value rather than prognostic significance.

Relationship between Plasma Levels of Lipopolysaccharide (LPS) and LPS-Binding Protein in Patients with Severe Sepsis and Septic Shock

Steven M. Opal, Patrick J. Scannon, Jean-Louis Vincent, Mark White, Stephen F. Carroll, John E. Palardy, Nicolas A. Parejo, John P. Pribble, Jon H. Lemke

The Journal of Infectious Diseases, Volume 180, Issue 5, November 1999, Pages 1584–1589, https://doi.org/10.1086/315093

Published: 01 November 1999 **Article history** ▾

Endotoxin (LPS) levels correlate with sepsis severity and mortality
Healthy: **<10 pg/mL** → Severe sepsis: **110–726 pg/mL (median ~300 pg/mL)** → Non-survivors: **~515 pg/mL**; endotoxemia present in **~78%** of severe sepsis cases, with levels **>50 pg/mL** indicating high-risk infection and increased mortality, especially in elderly and septic shock patients.

LBP reflects host inflammatory capacity, not endotoxin burden
Healthy: **~2–6 µg/mL** → Sepsis: **~10–60 µg/mL (median ~31 µg/mL)**; survivors show higher LBP (**~33 µg/mL**) than non-survivors (**~28 µg/mL**), indicating preserved hepatic acute-phase response; LBP and endotoxin levels are uncorrelated, representing independent biological axes.

Elevated Levels of Lipopolysaccharide-Binding Protein and Soluble CD14 in Plasma in Neonatal Early-Onset Sepsis

Authors: Reinhard Berner, Birgitt Fürll, Felix Stelter, Jana Dröse, Hans-Peter Müller, Christine Schütt | AUTHORS INFO & AFFILIATIONS

https://doi.org/10.1128/CDLI.9.2.440-445.2002 • Check for updates

LBP and sCD14 are significantly elevated at birth in neonatal sepsis and persist beyond 24 h
Septic neonates showed markedly higher plasma **LBP** (**median 36.6 µg/mL; IQR ~10.6–50.4 µg/mL**) compared to healthy controls (**median 7.8 µg/mL; IQR ~6.1–10.6 µg/mL; P < 0.001**), with levels further increasing at 24–48 h (**median ~60 µg/mL**); **sCD14** was also elevated (**median ~0.42 µg/mL vs 0.28 µg/mL in controls; P < 0.001**) but remained stable over time.

LBP and sCD14 reflect innate immune activation independent of pathogen type
Elevated **LBP (~20–60 µg/mL)** and **sCD14 (~0.35–0.65 µg/mL)** were observed regardless of blood culture positivity or Gram status (including *Streptococcus agalactiae*), and strongly correlated with pro-inflammatory cytokines (IL-6, IL-8, IL-1β, G-CSF), indicating activation of the neonatal innate immune response rather than Gram-negative–specific endotoxemia.

Perioperative change of plasma endotoxin levels in early infants

Kenji Imura [1], Yuichi Fukui [1], Makoto Yagi [1], Sumio Nakai [1], Toshimichi Hasegawa [1], Hisayoshi Kawahara [1], Shinkichi Kamata [1], Akira Okada [1]

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Endotoxin levels in immunocompromised children with fever

M.D. Ada Hass, B.S. Mark I. Rossberg, M.D. Horace L. Hodes, M.D. Alexander C. Hyatt, M.D. David S. Hodes

https://doi.org/10.1016/S0022-3476(86)80383-3

(abstract-derived information)

Defined upper limit of normal endotoxin in children

Plasma endotoxin levels in healthy and convalescent children were **≤35 pg/mL (0.10 EU/mL)**, establishing **35 pg/mL as the pediatric upper normal threshold** for endotoxemia.

Endotoxemia contributes to unexplained fever without bacteremia

14% (5/36) of immunocompromised children with febrile episodes showed **elevated endotoxin levels >35 pg/mL** despite absence of bacteremia or clinically diagnosed infection; endotoxin levels normalized during convalescence, implicating transient endotoxemia as a **potential pyrogenic mechanism** in unexplained pediatric fever.

Endotoxin-specific chromogenic assay for plasma in pregnant women, umbilical cords, neonates and children

Masako Nakajima MD [1], Masumi Inagaki MD [2], Yukinori Ando MD [2], Sachio Takashima MD [5 6], Kenzo Takeshita MD [2], Hiroyuki Tominaga MD [4], Shigenori Tanaka [1], Shuuji Shimizu MD [3]

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https://doi.org/10.1016/S0387-7604(88)80097-4

Physiological endotoxin levels are very low in normal pregnancy, neonates, and children

Using an endotoxin-specific chromogenic assay (Endospecy), endotoxin levels were **0.8 ± 0.4 pg/mL** in maternal blood and **1.3 ± 0.6 pg/mL** in umbilical cord blood after normal delivery; non-infectious neonates showed **~6.7 ± 5.1 pg/mL**, and healthy children had levels **<10 pg/mL**, defining a low baseline endotoxin range in pediatric populations.

Direct detection of bacteremia by exploiting host-pathogen interactions of lipoteichoic acid and lipopolysaccharide

Jessica Z. Kubicek-Sutherland, Dung M. Vu, Aneesa Noormohamed, Heather M. Mendez, Loreen R. Stromberg, Christine A. Pedersen, Astrid C. Hengartner, Katja E. Klosterman, Haley A. Bridgewater, Vincent Otieno, Qiuying Cheng, Samuel B. Anyona, Collins Ouma, Evans Raballah, Douglas J. Perkins, Benjamin H. McMahon & Harshini Mukundan

LPS and LTA are validated, pathogen-proximal biomarkers ideally suited for SERS detection
Direct detection of **lipopolysaccharide (LPS)** and **lipoteichoic acid (LTA)** captures conserved bacterial **PAMPs** that circulate in blood during bacteremia even at **very low bacterial loads (~1 CFU/mL)**; this supports **SERS-based, label-free detection** of molecular fingerprints rather than indirect host response markers or activity-dependent endotoxin assays.

Quantification of Lipoteichoic Acid in Hemodialysis Patients With Central Venous Catheters

Amy Barton Pai [1,*], Adinoyi Garba [2], Paul Neumann [3], Alexander J Prokopienko [4], Gabrielle Costello [1], Michael C Dean [1], Sriram Narsipur [5]

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PMCID: PMC6230575 PMID: 30456212

LTA is quantifiable systemically and reflects Gram-positive biofilm burden beyond culture detection
Lipoteichoic acid (LTA), a Gram-positive cell wall component, was detected in both catheter aspirate and plasma of hemodialysis patients with central venous catheters, with **plasma LTA 3.93 ng/mL (0.25–15)** exceeding **catheter aspirate LTA 2.38 ng/mL (0.1–8.1)** in **70% of patients ($p = 0.01$)**, demonstrating continuous biofilm shedding and systemic dissemination not captured by routine culture.

Functionalized arrays of Raman-enhancing nanoparticles for capture and culture-free analysis of bacteria in human blood

Ting-Yu Liu [1], Kun-Tong Tsai, Huai-Hsien Wang, Yu Chen, Yu-Hsuan Chen, Yuan-Chun Chao, Hsuan-Hao Chang, Chi-Hung Lin, Juen-Kai Wang, Yuh-Lin Wang

Affiliations + expand

PMID: 22086338 DOI: 10.1038/ncomms1546

Vancomycin-functionalized SERS substrates enable culture-free capture and direct Raman detection of whole bacteria from blood
Vancomycin-coated Ag nanoparticle arrays exploit specific hydrogen bonding with bacterial peptidoglycan, achieving up to **~1,000-fold enhancement in bacterial capture** while **excluding blood cells**, enabling **label-free, rapid SERS analysis directly from complex matrices like human blood**.

Chemical functionalization enhances SERS sensitivity and enables phenotypic discrimination
Vancomycin coating not only increases bacterial adhesion but also pulls bacterial cell walls into SERS “hot spots,” resulting in **400–500% higher Raman signal** and allowing **spectral differentiation between vancomycin-susceptible and resistant strains**, demonstrating SERS as a powerful platform for **rapid diagnostics and antimicrobial susceptibility assessment without culture**.

The biochemical origins of the surface enhanced Raman spectra of bacteria: metabolomics profiling by SERS

W Ranjith Premasiri [a,b,*], Jean C Lee [c], Alexis Sauer-Budge [b,d,e], Roger Theberge [f], Catherine E Costello [f], Lawrence D Ziegler [a,b,*]

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PMCID: PMC4911336 NIHMSID: NIHMS780783 PMID: 27100230

Bacterial SERS signals originate from extracellular purine metabolites, not cell-wall components
At 785 nm excitation, SERS spectra of whole bacteria are dominated by **purine degradation products**—adenine, hypoxanthine, xanthine, guanine, uric acid, and AMP—released during **starvation induced by water washing**. Identical SERS fingerprints from whole cells and cell-free supernatants confirm that **peptidoglycan (NAG/NAM), lipids, proteins, or LPS do not contribute significantly** to these spectra.

785 nm SERS functions as a metabolic stress readout rather than a structural biomarker
The bacterial SERS fingerprint reflects **species-specific purine metabolism and enzyme availability**, not Gram type or cell-envelope architecture. Signal intensity and composition are strongly influenced by **sample preparation, washing buffer, and starvation duration**, establishing SERS as a **metabolic biomarker platform sensitive to bacterial stress state**, rather than a direct detector of surface antigens or cell-wall molecules.

Escherichia coli research on Raman measurement mechanism and diagnostic model

Dongyu Ma [a], Xiaoyu Zhao [a], Chunjie Wang [a], Haoxuan Li [a], Yue Zhao [a], Lijing Cai [a],
Jinming Liu [a], Liang Tong [b]

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Raman/SERS signal origin is mechanistically linked to bacterial cell-wall chemistry

Raman/SERS signatures of *E. coli* can be **physically and biologically anchored to peptidoglycan biomarkers**, as Density Functional Theory (DFT)–validated vibrational modes of **N-acetylmuramic acid (MurNAc) and N-acetylglucosamine (GlcNAc)** directly explain the origin of diagnostic Raman bands, enabling **mechanism-based interpretation rather than empirical fingerprinting**.

Key Finding 2 — Mechanism-informed feature selection significantly enhances diagnostic accuracy

Incorporating **DFT-certified MurNAc/GlcNAc Raman peaks (3010–3029 cm^{-1})** with chemometric selection (SPA) and machine learning (Random Forest) **substantially reduces misdiagnosis** and achieves **near-clinical performance (~98% accuracy, AUC ≈ 1)**, demonstrating that **biomarker-guided Raman/SERS models outperform untargeted spectral classification**.

Comparison table for Which Solvent Has the *Highest Overall Solubility?*

Biomarker / Solvent	Water	PBS (pH 7.2–7.4)	DMSO	DMF	Ethanol
LTA (Staph. aureus)	5 mg/mL (hazy) (MilliporeSigma)	Moderate (hazy; comparable to water), *qualitative* (MilliporeSigma)	Unknown	Unknown	Moderate (MilliporeSigma)
MurNAc	~50 mg/mL (clear) (ChemicalBook)	~10 mg/mL (Cayman Chemical)	~30 mg/mL (Cayman Chemical)	~2.5 mg/mL (Cayman Chemical)	Unknown
LPS (E. coli O111:B4)	~5 mg/mL (hazy; up to ~20 mg/mL with heat) (MilliporeSigma)	Moderate (hazy solutions in PBS; similar to water) (MilliporeSigma)	Poor / Not reported (organic solvents do not yield clearer solutions) (MilliporeSigma)	Unknown	Poor (methanol/ethanol produce turbid suspensions with precipitates) (MilliporeSigma)

Best Solvent for Each Biomarker

Biomarker	Best Solvent /Highest Solubility
LTA	Water (5 mg/mL)
MurNAc	Water (~50 mg/mL)
LPS	Water (~5 mg/mL)